AdventureWorks Sales Analytics

A SQL Server Data Analysis Project

Prepared by Azeem Khan — Data Analyst

Database: AdventureWorks2025 (database version) | Order data: 30 May 2022 – 29 Jun 2025 | 31,465 orders | 20 queries

$123.2M	31,465	19,119	10
Total revenue (TotalDue)	Orders analyzed	Customers with orders	Sales territories

Executive Summary

This project analyzes three years of order data (May 2022 – June 2025) from AdventureWorks, a fictional bicycle manufacturer, using Microsoft SQL Server. It answers questions a Sales Operations or BI team would ask: which regions and products drive revenue, how reseller and individual customers differ, and which customers are at risk of churning. Beyond one-off queries it packages a parameterized stored procedure and two views designed to connect to Power BI or Excel.

Note on years: “2025” in AdventureWorks2025 is the database version, not the data period. The orders themselves run from 30 May 2022 to 29 Jun 2025, so 2022 and 2025 are partial years.

What This Project Demonstrates

- Multi-table SQL joins (up to 4 tables per query), aggregation and CTEs
- Window functions: RANK / DENSE_RANK, LAG, running totals
- Turning raw data into business insights
- Reusable database objects: stored procedure and views
- Cross-checking results before reporting them

Tools Used

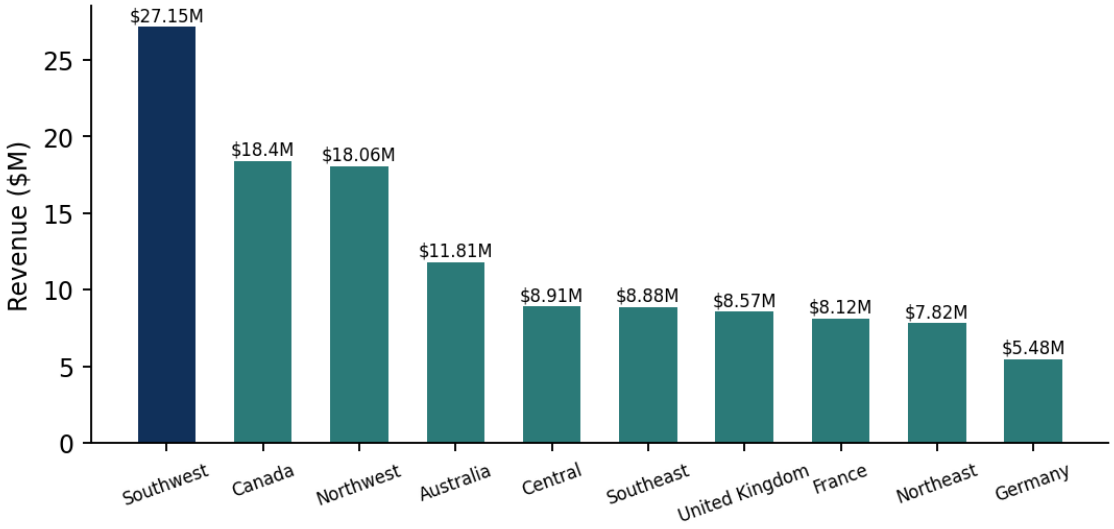
SQL Server / SSMS (version: confirm with SELECT @@VERSION) · T-SQL · AdventureWorks2025 sample OLTP database · Power BI / Excel (planned front-end)

Approach

1. Explore – data volume, date range and NULL checks. 2. Answer business questions – region, product, category, customer value. 3. Advanced techniques – growth rates, ranking within groups, churn, segmentation. 4. Package – stored procedure and two views. 5. Verify – views and queries were cross-checked against each other; *05_validation_checks.sql* re-tests the key claims.

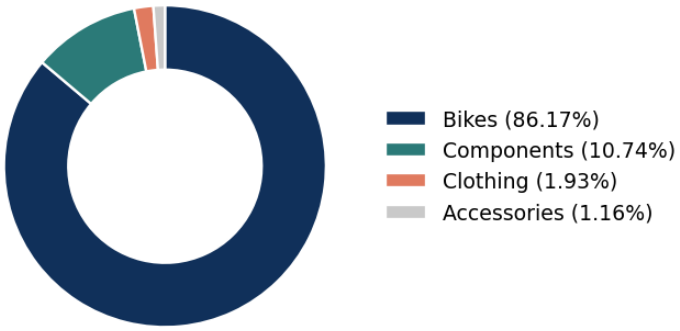
Key Findings

1. Revenue is concentrated in a few territories



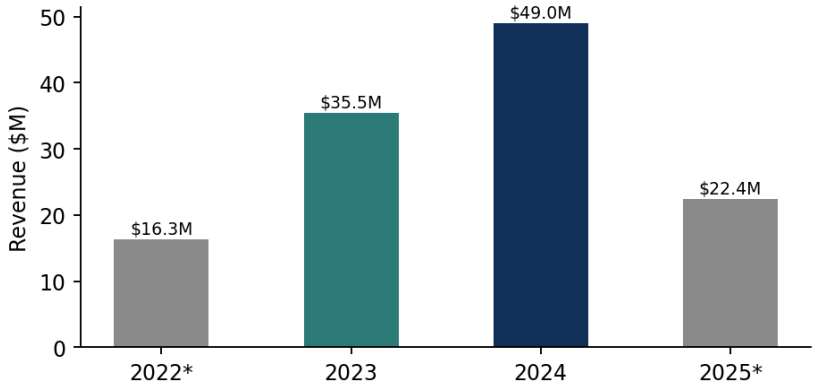
Southwest leads at $27.15M, about 48% ahead of Canada ($18.4M). Germany is lowest at $5.48M, but low revenue is not automatically underperformance: Central, Northeast and Southeast earn ~$8M from only 350–490 large reseller orders each (AOV ≈ $18–23K), while Germany has 2,623 orders at ~$2K AOV. Territories should be compared on AOV and customer mix, not revenue alone.

2. The business depends on one product category



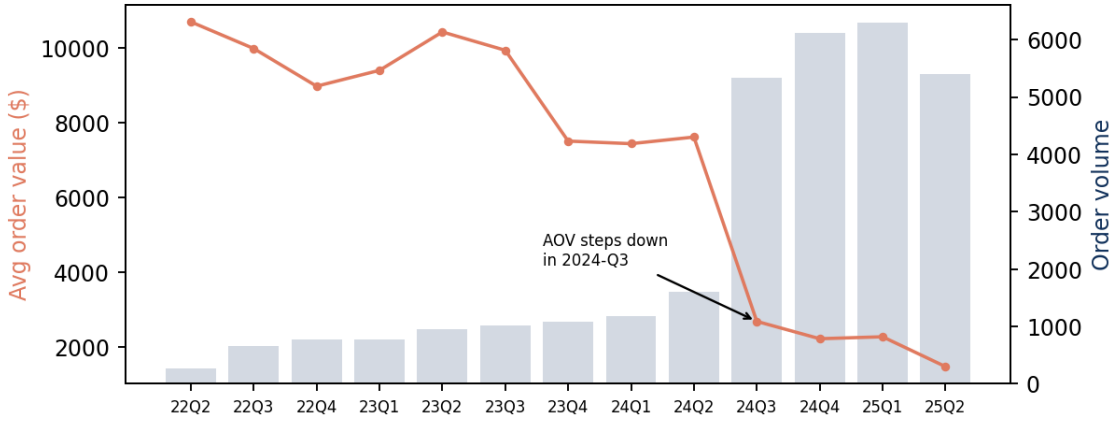
Bikes account for **86.17% of product sales** ($94.65M of $109.85M line-item sales, before tax and freight). Components are 10.74%; Clothing and Accessories together about 3%. Note: the $123.2M headline revenue uses TotalDue (includes tax and freight), so category shares are quoted as % of product sales. Any softening in bike demand would hit hard.

3. Revenue grew in 2024, but growth slowed in 2025



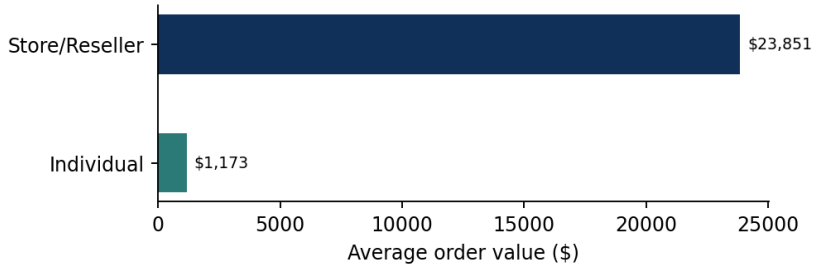
2023 → 2024 revenue grew +38% ($35.5M → $49.0M). 2022 covers only 30 May – Dec, so comparing it to 2024 (“3x”) is not like-for-like. Comparing the same half-years, H1 2025 (~$22.4M) is only ~6% above H1 2024 (~$21.0M, derived from quarterly figures) even though order count roughly quadrupled.

4. Order volume surged, but average order value dropped in a step



Quarterly orders grew from 264 (2022-Q2) to 5,402 (2025-Q2), a 20x rise. AOV stayed around $7,500+ until 2024-Q2, then fell to ~$2,700 in 2024-Q3 and ~$1,480 by 2025-Q2. This is consistent with growth coming from high-volume individual buyers instead of large reseller orders; the quarter × customer-type query in 05_validation_checks.sql confirms or rejects this.

5. Reseller customers are worth far more per order



Store/Reseller customers average $23,850 per order versus $1,173 for individuals (~20x). With 3,806 reseller orders against 27,659 individual orders, the two segments should be reported and targeted separately.

6. A meaningful share of customers have gone quiet

1,075 customers (about 5.6% of the 19,119 customers who placed orders) have not ordered in 12+ months, measured against the latest date in the data (not today's date). The most dormant customer last ordered on 30 May 2022 (37 calendar months before the last order date). This is a ready-made win-back list.

7. Results were cross-checked

The sales-summary view returned exactly 31,465 rows (matching the raw order count); the product-performance view's top product (Mountain-200 Black, 38: $4.40M) matched the raw top-seller query; the segmentation's top customer (ID 29818, $989,184) matched the top-spender query.

Recommendations

• Compare territories by AOV and reseller/individual mix before labelling any as underperforming.
• Report Reseller and Individual customers separately.
• Launch a win-back campaign for the 1,075 dormant customers, prioritized by past spend.
• Investigate the 2024-Q3 drop in AOV and why revenue growth slowed in H1 2025 despite much higher order volume.
• Connect Power BI / Excel to vw_SalesSummary and vw_ProductPerformance instead of raw tables.

Project Artifacts

20 queries in total: 17 in four scripts (01–04), plus a stored procedure and two views, along with a validation script (05) and a query-by-query revision guide on GitHub.

Azeem Khan | Data Analyst | azeemkhanmev2@gmail.com | linkedin.com/in/azeem--khan | github.com/AzeemDataAnalyst